# TITRATIONS 3

**1** 25.0 $cm^3$ of 0.200 mol/$dm^3$ sulfuric acid neutralises 18.6 $cm^3$ of barium hydroxide solution.

$$H_2SO_4(aq) + Ba(OH)_2(aq) \rightarrow BaSO_4(s) + 2\ H_2O(l)$$

**a** Find the concentration of the barium hydroxide solution in mol/$dm^3$. Give your answer to 3 significant figures.

**moles $H_2SO_4$ = conc x vol ($dm^3$) = 0.200 x $\frac{25.0}{1000}$ = 0.00500 mol**

**moles $Ba(OH)_2$ = moles of $H_2SO_4$ = 0.00500 mol**

**conc $Ba(OH)_2$ = $\frac{\text{moles}}{\text{volume (dm}^3\text{)}}$ = $\frac{0.00500}{\frac{18.6}{1000}}$ = 0.269 mol/$dm^3$**

**b** Find the concentration of the barium hydroxide solution in g/$dm^3$. Give your answer to 3 significant figures.

**conc $Ba(OH)_2$ = 171 x 0.269 = 46.0 g/$dm^3$**

**c** List the key steps in the practical you would carry out to find out that 18.6 $cm^3$ of the barium hydroxide solution is needed to neutralise 25.0 $cm^3$ of 0.200 mol/$dm^3$ sulfuric acid.

**use a pipette to place 25 $cm^3$ of sulfuric acid**
**In a conical flask**
**add an indicator**
**add barium hydroxide from burette**
**until colour changes**
**go dropwise near end-point**
**record result**
**repeat**

**2** What volume of 0.0100 mol/$dm^3$ calcium hydroxide neutralises 25.0 $cm^3$ of 0.140 mol/$dm^3$ sulfuric acid? Give your answer to 3 significant figures

$$H_2SO_4(aq) + Ca(OH)_2(aq) \rightarrow CaSO_4(s) + 2\ H_2O(l)$$

**moles $H_2SO_4$ = conc x vol ($dm^3$) = 0.140 x $\frac{25.0}{1000}$ = 0.00350 mol**

**moles $Ca(OH)_2$ = moles $H_2SO_4$ = 0.00350 mol**

**volume $Ca(OH)_2$ = $\frac{\text{moles}}{\text{conc}}$ = $\frac{0.00350}{0.0100}$ = 0.350 $dm^3$**

**3** What volume of 1.50 mol/$dm^3$ potassium hydroxide reacts with 25.0 $cm^3$ of 1.10 mol/$dm^3$ nitric acid? Give your answer to 3 significant figures

$$KOH(aq) + HNO_3(aq) \rightarrow KNO_3(aq) + H_2O(l)$$

**moles $HNO_3$ = conc x vol ($dm^3$) = 1.10 x $\frac{25.0}{1000}$ = 0.0275 mol**

**moles KOH = moles $HNO_3$ = 0.0275 mol**

**volume KOH = $\frac{\text{moles}}{\text{conc}}$ = $\frac{0.0275}{1.50}$ = 0.0183 $dm^3$**

**4** 25.0 $cm^3$ of sodium carbonate solution neutralises 27.9 $cm^3$ of 0.500 $mol/dm^3$ nitric acid solution.

$$2\ HNO_3(aq) + Na_2CO_3(aq) \rightarrow 2\ NaNO_3(aq) + CO_2(aq) + 2\ H_2O(l)$$

**a** Find the concentration of sodium carbonate solution in $mol/dm^3$. Give your answer to 3 significant figures

**moles $HNO_3$ = conc x vol ($dm^3$) = 0.500 x $\frac{27.9}{1000}$ = 0.01395 mol**

**moles $Na_2CO_3$ = $\frac{1}{2}$ x moles of $HNO_3$ = $\frac{1}{2}$ x 0.01395 = 0.006975 mol**

**conc $Na_2CO_3$ = $\frac{\text{moles}}{\text{volume } (dm^3)}$ = $\frac{0.006975}{\frac{25.0}{1000}}$ = 0.279 $mol/dm^3$**

**b** Find the concentration of the sodium carbonate solution in $g/dm^3$. Give your answer to 3 significant figures

**conc $Na_2CO_3$ = 106 x 0.279 = 29.6 $g/dm^3$**

**5** A 0.1575 g sample of ethanedioic acid crystals, $H_2C_2O_4.nH_2O$ was dissolved in water. In a titration, 25.0 $cm^3$ of this solution of acid reacted with exactly 15.6 $cm^3$ of 0.160 $mol/dm^3$ NaOH. Calculate the value of $M_r$ of the acid and n.

$$H_2C_2O_4.nH_2O(aq) + 2\ NaOH(aq) \rightarrow Na_2C_2O_4(aq) + (2+n)\ H_2O(l)$$

**moles NaOH = conc x vol ($dm^3$) = 0.160 x $\frac{15.6}{1000}$ = 0.002496 mol**

**moles $H_2C_2O_4.nH_2O$ = $\frac{1}{2}$ x moles NaOH = $\frac{1}{2}$ x 0.002496 = 0.001248 mol**

**$M_r$ $H_2C_2O_4.nH_2O$ = $\frac{\text{mass}}{\text{moles}}$ = $\frac{0.1575}{0.001248}$ = 126**

**$M_r$ $nH_2O$ = 126 – [ 2(1) + 2(12) + 4(16) ] = 36**

**n = $\frac{36}{18}$ = 2**

| Area | Strength | To develop | Area | Strength | To develop | Area | Strength | To develop |
|---|---|---|---|---|---|---|---|---|
| Done with care and thoroughness | | | Can find conc or vol of other reagent | | | Can describe how to do a titration | | |
| Shows suitable working | | | Can convert $mol/dm^3$ to $g/dm^3$ | | | Names correct apparatus | | |
| Can work out moles from conc & vol | | | Does not round too much | | | Gives units | | |
| Uses equation for other reactant moles | | | Can use sig figs | | | | | |